

Machine Learning 1

Mock Exam A – Answer Key

Worked solutions for all 100 points

Equivalent algebraic forms are valid. The derivations below use numerator-layout derivatives where a scalar derivative is a row vector.

Exercise 1: Multivariate derivatives and regularization

1. Temperature-scaled log-softmax

Answer. Since

$$\ell_i(x) = \frac{x_i}{\tau} - \log \left(\sum_{k=1}^d e^{x_k/\tau} \right),$$

we obtain

$$\begin{aligned} \frac{\partial \ell_i}{\partial x_j} &= \frac{\delta_{ij}}{\tau} - \frac{e^{x_j/\tau}/\tau}{\sum_{k=1}^d e^{x_k/\tau}} \\ &= \frac{1}{\tau}(\delta_{ij} - s_j). \end{aligned}$$

2. Full Jacobian

Answer. Row i , column j must contain $(\delta_{ij} - s_j)/\tau$. Therefore

$$J_\ell(x) = \frac{1}{\tau} \left(I_d - \mathbf{1} s(x)^\top \right) \in \mathbb{R}^{d \times d}.$$

The matrix $\mathbf{1} s^\top$ has entry s_j in every row.

3. Invariant direction

Answer. Since $s^\top \mathbf{1} = 1$,

$$J_\ell(x) \mathbf{1} = \frac{1}{\tau} \left(\mathbf{1} - \mathbf{1} s^\top \mathbf{1} \right) = 0.$$

For any scalar c ,

$$s_i(x + c\mathbf{1}) = \frac{e^{c/\tau} e^{x_i/\tau}}{e^{c/\tau} \sum_k e^{x_k/\tau}} = s_i(x).$$

Thus $\ell(x + c\mathbf{1}) = \ell(x)$. The directional derivative along $\mathbf{1}$ is consequently zero.

4. Generalized ridge regression

Answer. The numerator-layout derivative is

$$\frac{\partial L}{\partial \theta} = \frac{2}{n} (X\theta - y)^\top X + 2\gamma \theta^\top R.$$

After transposing the stationarity equation,

$$\frac{1}{n} X^\top (X\theta - y) + \gamma R\theta = 0,$$

so

$$(X^\top X + n\gamma R)\theta = X^\top y, \quad \boxed{\hat{\theta} = (X^\top X + n\gamma R)^{-1} X^\top y}.$$

For any nonzero $v \in \mathbb{R}^p$,

$$v^\top (X^\top X + n\gamma R)v = \|Xv\|_2^2 + n\gamma v^\top Rv > 0.$$

Thus the matrix is positive definite even if X is rank deficient. Equivalently, the Hessian

$$\frac{2}{n} X^\top X + 2\gamma R$$

is positive definite. Hence L is strictly convex and the stationary point is the unique global minimizer.

Exercise 2: Waiting times and Bayesian estimation

1. Expected waiting time

Answer. Put $r = 1 - q$. Differentiating the geometric series gives

$$\sum_{k=1}^{\infty} k r^{k-1} = \frac{1}{(1-r)^2}, \quad \sum_{k=0}^{\infty} k r^k = \frac{r}{(1-r)^2}.$$

Therefore

$$\mathbb{E}[K \mid q] = q \sum_{k=0}^{\infty} k(1-q)^k = q \frac{1-q}{q^2} = \boxed{\frac{1-q}{q}}.$$

2. Requested probability

Answer.

$$P(K = 2 \mid q = 0.25) = 0.25(0.75)^2 = \boxed{\frac{9}{64} = 0.140625}.$$

3. Likelihood and log-likelihood

Answer. By independence,

$$L(q) = \prod_{n=1}^N q(1-q)^{K_n} = q^N (1-q)^S.$$

Thus

$$\boxed{\log L(q) = N \log q + S \log(1-q)}$$

up to additive constants independent of q .

4. Maximum-likelihood estimator

Answer.

$$\frac{d}{dq} \log L(q) = \frac{N}{q} - \frac{S}{1-q} = 0 \quad \implies \quad \boxed{\hat{q}_{\text{ML}} = \frac{N}{N+S}}.$$

Because $S > 0$, this value is in $(0, 1)$. Also,

$$\frac{d^2}{dq^2} \log L(q) = -\frac{N}{q^2} - \frac{S}{(1-q)^2} < 0,$$

so the log-likelihood is strictly concave and this is its unique maximum.

5. Posterior distribution

Answer. Multiplication of the likelihood and prior gives

$$\begin{aligned} p(q \mid K_1, \dots, K_N) &\propto q^N (1-q)^S q^{a-1} (1-q)^{b-1} \\ &= q^{a+N-1} (1-q)^{b+S-1}. \end{aligned}$$

Therefore

$$q \mid K_1, \dots, K_N \sim \text{Beta}(a + N, b + S).$$

6. MAP estimator

Answer. The derivative of the posterior log-density is

$$\frac{a + N - 1}{q} - \frac{b + S - 1}{1 - q}.$$

Setting it to zero gives

$$\hat{q}_{\text{MAP}} = \frac{a + N - 1}{a + b + N + S - 2}.$$

Both posterior shape parameters exceed one, so the mode is unique and interior.

7. Data statistic

Answer. The statistic is

$$S = \sum_{n=1}^N K_n.$$

For fixed N , both the likelihood and posterior depend on the observed values only through S . Thus S is sufficient for q by the factorization criterion.

Exercise 3: Estimating an unknown range

1. Likelihood

Answer. Each observation has density

$$p(x_n \mid M) = \frac{1}{2M} \mathbf{1}\{|x_n| \leq M\}.$$

Consequently,

$$p(x_1, \dots, x_N \mid M) = (2M)^{-N} \mathbf{1}\{M \geq A\}, \quad M > 0.$$

2. Maximum-likelihood estimator

Answer. The likelihood is zero for $M < A$ and equals $(2M)^{-N}$ for $M \geq A$. It is strictly decreasing over the feasible range, so

$$\hat{M}_{\text{ML}} = A = \max_n |X_n|.$$

For the supplied observations,

$$A = \max(3.2, 1.1, 2.4, 2.8, 0.5) = \boxed{3.2}.$$

3. Unbiased correction

Answer. For $\widetilde{M} = cA$, unbiasedness requires

$$\mathbb{E}[\widetilde{M}] = c \frac{NM}{N+1} = M.$$

Thus

$$\boxed{\widetilde{M} = \frac{N+1}{N}A}.$$

4. Variance

Answer. First,

$$\begin{aligned} \text{Var}(A) &= \frac{NM^2}{N+2} - \frac{N^2M^2}{(N+1)^2} \\ &= \frac{NM^2}{(N+2)(N+1)^2}. \end{aligned}$$

Therefore

$$\boxed{\text{Var}(\widetilde{M}) = \left(\frac{N+1}{N}\right)^2 \text{Var}(A) = \frac{M^2}{N(N+2)}}.$$

Exercise 4: Multi-output regression with correlated noise

1. Dimensions

Answer.

$$\boxed{\Phi W \in \mathbb{R}^{N \times k}}, \quad \boxed{T - \Phi W \in \mathbb{R}^{N \times k}}, \quad \boxed{(T - \Phi W)^\top (T - \Phi W) \in \mathbb{R}^{k \times k}}.$$

2. Log-likelihood

Answer. Let $E(W) = T - \Phi W$. Since

$$|\sigma^2 C| = (\sigma^2)^k |C|,$$

independence gives

$$\begin{aligned} \log p(T \mid W, \sigma^2) &= -\frac{Nk}{2} \log(2\pi) - \frac{Nk}{2} \log \sigma^2 - \frac{N}{2} \log |C| \\ &\quad - \frac{1}{2\sigma^2} \text{tr} \left(C^{-1} E(W)^\top E(W) \right). \end{aligned}$$

3. Maximum-likelihood estimator of W

Answer. For fixed σ^2 , minimize

$$Q(W) = \text{tr}(C^{-1} E^\top E).$$

Using $dE = -\Phi dW$ and symmetry of C^{-1} ,

$$dQ = -2 \text{tr}(C^{-1} E^\top \Phi dW),$$

so stationarity gives

$$\Phi^\top (T - \Phi W) C^{-1} = 0.$$

Since C is invertible,

$$\Phi^\top \Phi W = \Phi^\top T, \quad \boxed{W_{\text{ML}} = (\Phi^\top \Phi)^{-1} \Phi^\top T}.$$

The positive scalar $1/\sigma^2$ cancels, and multiplication by the invertible C^{-1} does not change the normal equations. Hence the estimate depends on neither C nor σ^2 .

4. Maximum-likelihood estimator of σ^2

Answer. Let $E = T - \Phi W_{\text{ML}}$ and

$$Q = \text{tr}(C^{-1} E^\top E) > 0.$$

For $z = \sigma^2$, the relevant log-likelihood is

$$-\frac{Nk}{2} \log z - \frac{Q}{2z}.$$

Its derivative is $-Nk/(2z) + Q/(2z^2)$. Therefore

$$\boxed{\hat{\sigma}_{\text{ML}}^2 = \frac{\text{tr}(C^{-1} E^\top E)}{Nk}}.$$

At this value the second derivative is $-Nk/(2z^2) < 0$. The nonzero-residual assumption ensures a positive estimate and an attained maximum.

5. Residual orthogonality

Answer. The normal equations imply

$$\boxed{\Phi^\top E = \Phi^\top (T - \Phi W_{\text{ML}}) = 0}.$$

Thus every residual column is orthogonal to every column of Φ . For each output, the fitted vector is the Euclidean orthogonal projection of the target onto the column space of Φ .

Exercise 5: Information from one Bayesian observation

Write

$$D = \beta^{-1} + \phi^\top \Sigma_N \phi > 0.$$

1. Covariance update

Answer. Apply Sherman–Morrison to

$$\Sigma_{N+1} = (\Sigma_N^{-1} + \beta \phi \phi^\top)^{-1}.$$

This gives

$$\begin{aligned} \Sigma_{N+1} &= \Sigma_N - \frac{\beta \Sigma_N \phi \phi^\top \Sigma_N}{1 + \beta \phi^\top \Sigma_N \phi} \\ &= \boxed{\Sigma_N - \frac{\Sigma_N \phi \phi^\top \Sigma_N}{\beta^{-1} + \phi^\top \Sigma_N \phi}}. \end{aligned}$$

2. Innovation form of the mean

Answer. Start from

$$\mu_{N+1} = \Sigma_{N+1}(\Sigma_N^{-1}\mu_N + \beta\phi t).$$

The covariance update implies

$$\Sigma_{N+1}\Sigma_N^{-1} = I - \frac{\Sigma_N\phi\phi^\top}{D}, \quad \Sigma_{N+1}\beta\phi = \frac{\Sigma_N\phi}{D}.$$

Hence

$$\boxed{\mu_{N+1} = \mu_N + K(t - \phi^\top \mu_N)}, \quad \boxed{K = \frac{\Sigma_N\phi}{D}}.$$

The vector $\Sigma_N\phi$ selects the parameter direction correlated with the measured quantity $\phi^\top w$. The scalar innovation $t - \phi^\top \mu_N$ gives the signed prediction error and controls the update magnitude.

3. Posterior predictive distribution

Answer. Write $t = \phi^\top w + \varepsilon$, where $w \mid \mathcal{D}_N \sim \mathcal{N}(\mu_N, \Sigma_N)$ and $\varepsilon \sim \mathcal{N}(0, \beta^{-1})$ are independent. Therefore

$$\boxed{p(t \mid \phi, \mathcal{D}_N) = \mathcal{N}\left(t \mid \phi^\top \mu_N, \beta^{-1} + \phi^\top \Sigma_N \phi\right)}.$$

The variance consists of observation-noise uncertainty β^{-1} and posterior parameter uncertainty $\phi^\top \Sigma_N \phi$.

4. Directional uncertainty reduction

Answer. From the covariance update,

$$\begin{aligned} a^\top (\Sigma_N - \Sigma_{N+1}) a &= \frac{a^\top \Sigma_N \phi \phi^\top \Sigma_N a}{D} \\ &= \boxed{\frac{(a^\top \Sigma_N \phi)^2}{\beta^{-1} + \phi^\top \Sigma_N \phi}} \geq 0. \end{aligned}$$

It is zero exactly when $a^\top \Sigma_N \phi = 0$. This quantity is the posterior covariance between $a^\top w$ and $\phi^\top w$. Thus the observation reduces uncertainty only in parameter directions correlated with the measured quantity; the rank-one reduction acts along $\Sigma_N \phi$.

End of Mock Exam A Answer Key